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Next item →

1. (Special note: Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file [here](#) [🔗](#). If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.)

1 / 1 point

Consider the following LP

$$\begin{array}{ll} \max & 2x_1 - x_2 \\ \text{s.t.} & 8x_1 - 4x_2 \leq 16 \\ & 3x_1 - 4x_2 \leq 12 \\ & x_1 \geq 0, x_2 \leq 0. \end{array}$$

Determine whether it has a unique optimal solution, has multiple optimal solutions, is infeasible, or is unbounded. Moreover, list the binding constraints, if any, at $(x_1, x_2) = (2, 0)$.

☐ The LP is unbounded. The constraints $8x_1 - 4x_2 \leq 16$ and $x_2 \leq 0$ are binding at $(x_1, x_2) = (2, 0)$.

☐ The LP is unbounded. The constraint $8x_1 - 4x_2 \leq 16$ is binding at $(x_1, x_2) = (2, 0)$.

☒ The LP has multiple optimal solution. The constraints $8x_1 - 4x_2 \leq 16$ and $x_2 \leq 0$ are binding at $(x_1, x_2) = (2, 0)$.

☐ The LP has multiple optimal solution. The constraint $8x_1 - 4x_2 \leq 16$ is binding at $(x_1, x_2) = (2, 0)$.

☐ The LP has a unique optimal solution. The constraints $8x_1 - 4x_2 \leq 16$ and $x_2 \leq 0$ are binding at $(x_1, x_2) = (2, 0)$.

✔ Correct

2. Consider the following true/false questions:

1 / 1 point

1. If an LP is unbounded, its feasible region must be unbounded.

2. If an LP has an unbounded feasible region, it must be unbounded.

3. If an LP has an optimal solution, there must be at least two binding constraint(s) at that optimal solution.

4. If an LP has two optimal solutions, there must be another optimal solution that is different from the first two.

5. An LP's optimal solution is always an extreme point.

Provide your answers in the text box below with five consecutive uppercase "T" or "F". For example, if you believe the answers should be false, false, true, false, and true, type "FFTFT".

TFFTF

✔ Correct

3. Graphically solve the following LP:

1 / 1 point

$$\begin{array}{ll} \max & 3x_1 + 5x_2 \\ \text{s.t.} & x_1 + x_2 \leq 16 \\ & x_1 + 4x_2 \leq 20 \\ & 2x_1 + x_2 \geq 6 \\ & x_1 \geq 0, x_2 \geq 0. \end{array}$$

Which of the following is an optimal solution?

☐ (16, 0).

☒ $(\frac{44}{3}, \frac{4}{3})$.

☐ $(\frac{4}{7}, \frac{34}{7})$.

☐ (3, 0).

☐ None of the above.

✔ Correct

4. Tom is the owner of a furniture shop. He uses woods to make tables and chairs. Each day, he buys woods from his supplier at a cost of \$13 per unit. The maximum amount of woods that may be purchased is 50 units. Each table requires 7 units of woods while each chair requires 4 units. He needs to spend time on making these products. He can make 0.7 chair or 1.1 table in 1 hour. The outputs are always proportional to the amount of time they spend. Tom can work 10 hours per day. A table can be sold at \$100 and a chair can be sold at \$70.

1 / 1 point

Let x_1 and x_2 be the numbers of table and chair produced. A formulate an LP that can help make a production plan for Tom to maximize his average daily profit is

$$\begin{array}{ll} \max & Ax_1 + 18x_2 \\ \text{s.t.} & 7x_1 + 4x_2 \leq 50 \\ & Bx_1 + Cx_2 \leq 10 \\ & x_i \geq 0 \quad \forall i = 1, 2. \end{array}$$

However, three numbers are missing (labeled as A , B , and C). What should be the number to make the formulation correct?

☐ $A = 12, B = 1.1$, and $C = 0.7$.

☐ $A = 9, B = 1.1$, and $C = 0.7$.

☐ $A = 12, B = \frac{1}{1.1}$, and $C = \frac{1}{0.7}$.

☒ $A = 9, B = \frac{1}{1.1}$, and $C = \frac{1}{0.7}$.

☐ None of the above.

✔ Correct

5. Following from the above problem, graphically solve the LP and find an optimal solution. What is the optimal number of chairs to produce in each day?

1 / 1 point

7

✔ Correct